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Multicenter, retrospective study of balloon valvuloplasty-associated pulmonary artery dissection in dogs with pulmonary valve stenosis☆,☆☆

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Abstract *Introduction/objectives:* Pulmonary artery dissection (PAD) is a rare complication in dogs with pulmonary valve stenosis (PS) that undergo balloon valvuloplasty (BVP). This study aimed to describe demographic, interventional, and survival data in a cohort of dogs with PS that experienced PAD during BVP.
Animals, Materials and Methods: Fifty-three dogs were included in this retrospective, multicenter study, with medical records and available stored images reviewed.
Results: Dogs that developed PAD during BVP had a median body weight of 12.5 kg (range: 1.1—50.7 kg), median age of 0.8 years (range: 0.2—8.1 years), and 30/53 (56.6%) were male. Pit Bull terriers, Pit Bull terrier mixes, and 'Bully breeds' represented 23 of 53 dogs in the study (43.4%). The median angiographic balloon diameter to pulmonary valve annulus ratio of the largest balloon utilized during BVP was 1.4 (range: 1.0—1.58). High-pressure balloons (>6 atm) were utilized in 27 dogs (50.9%). Post-BVP transthoracic echocardiographic imaging clearly demonstrated the PAD in 62.2% of dogs, while 90.3% and 84.0% of available post-BVP angiography and transesophageal echocardiographic studies did, respectively. Forty-eight dogs (90.6%) survived to discharge. Median time from BVP to last follow-up or death (if applicable) was 2.0 years (range: 0.2—10.5 years), with all 27 dogs for whom ≥1 year of follow-up data were available confirmed alive at that time.
Study limitations: Retrospective study design.
Conclusions: Dogs experiencing PAD during BVP have a high likelihood of survival in the immediate and long-term periods. Immediate post-BVP imaging with angiography or transesophageal echocardiography appears more sensitive than transthoracic echocardiography for diagnosis of PAD.
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Abbreviations

BVP	balloon valvuloplasty
LV	left ventricle
PAD	pulmonary artery dissection
PS	pulmonary valve stenosis
PT	pulmonary trunk
RV	right ventricle
STJ	sinotubular junction
TEE	transesophageal echocardiography
TTE	transthoracic echocardiography

Introduction

Pulmonary artery dissection (PAD) is a rare condition stemming from a tear through the inner, intimal layer of the pulmonary arterial wall into the mid or deep medial layer. Separation of the vessel wall layers with extravasated blood results in dissection of the vessel and creates a false lumen within the artery. Due to the relatively thinner tunica media layer in the pulmonary artery compared to the aorta, PAD is considered more likely to result in vessel rupture compared to aortic dissection, with a high mortality rate historically reported in humans [1—5].

The primary cause of PAD in humans is chronic pulmonary hypertension, with pulmonary hypertension secondary to underlying congenital heart disease being a leading cause [5]. Patent ductus arteriosus is the most common congenital defect documented in association with spontaneous PAD in humans, followed by other shunting lesions such as ventricular septal defect, atrial septal defect or aortopulmonary window [5]. In dogs, PAD has been reported in conjunction with congenital shunting lesions including patent ductus arteriosus [6,7] and an anomalous vessel between the ascending aorta and pulmonary trunk (PT) [8].

In contrast, dogs and children with pulmonary valve stenosis (PS) are reported to develop PAD as a rare complication secondary to balloon valvuloplasty (BVP) intervention [9—14]. At present, there are only seven dogs described in the literature with PS that developed PAD during BVP. In a case series of six dogs, perioperative mortality was 50%, with the remaining 50% of dogs experiencing extended survival times after discharge (>3 years) [10]. In the other case report, only short-term survival was available, but the dog survived to discharge and had no reported clinical signs at a 2-month recheck [9].

Given the rarity of PAD in dogs, specifically in dogs undergoing BVP for PS, minimal data are available to draw conclusions about possible demographic predilections, or the impact of catheterization equipment choice on the likelihood of a PAD outcome. Additionally, survival data are not well understood. The objective of the present study was to gather a larger cohort of dogs with PS that experienced PAD during BVP and describe demographic, interventional, and survival data.

Animals, materials, and methods

Medical records and available stored images were reviewed from dogs with echocardiographic, angiographic, computed tomographic, or post-mortem diagnoses of PAD that developed during BVP intervention for treatment of PS. Cases were identified from 19 different veterinary hospitals located in the United States, Brazil, Germany and Italy. To be included, pre-procedural transthoracic echocardiography (TTE), angiography, and/or transesophageal echocardiography (TEE) had to have confirmed the absence of pre-existing PAD prior to the BVP procedure.

Data collected from the medical records included demographic data, BVP procedural date, heart murmur, reported pre-operative clinical signs, presence or absence of right-sided congestive heart failure and pre-procedural cardiac medications.

Pre-procedural TTE images were reviewed, if available, and the following variables were collected by contributing authors: subjective assessment of right atrial enlargement (none, mild, moderate, or severe); degree of concentric hypertrophy of the right ventricle as defined based on a comparison of the right ventricular free wall thickness at end-diastole to the left ventricle (LV) free wall thickness at end-diastole: mild = 0.5—1x LV, moderate = 1—1.5× LV, severe = >1.5× LV [10]; maximum transpulmonary velocity and corresponding pressure gradient; pulmonary valve anatomy features (characterized as feature 1 = dysplastic, thickened leaflets, feature 2 = tethered or doming leaflets, and/or feature 3 = annular hypoplasia; Fig. 1); pulmonary stenosis lesion localization(s) (valvular: anatomic abnormalities associated with valve leaflet size, thickness or motion and/or annular size; sinotubular junction (STJ): STJ-a = thickened band or bulge of tissue at the STJ, STJ-b = pulmonary valve leaflets adhered to STJ; supravalvular: discrete lesion geographically distant from the annulus and STJ; subvalvular: subvalvular-fixed = discrete lesion geographically distant from the annulus,

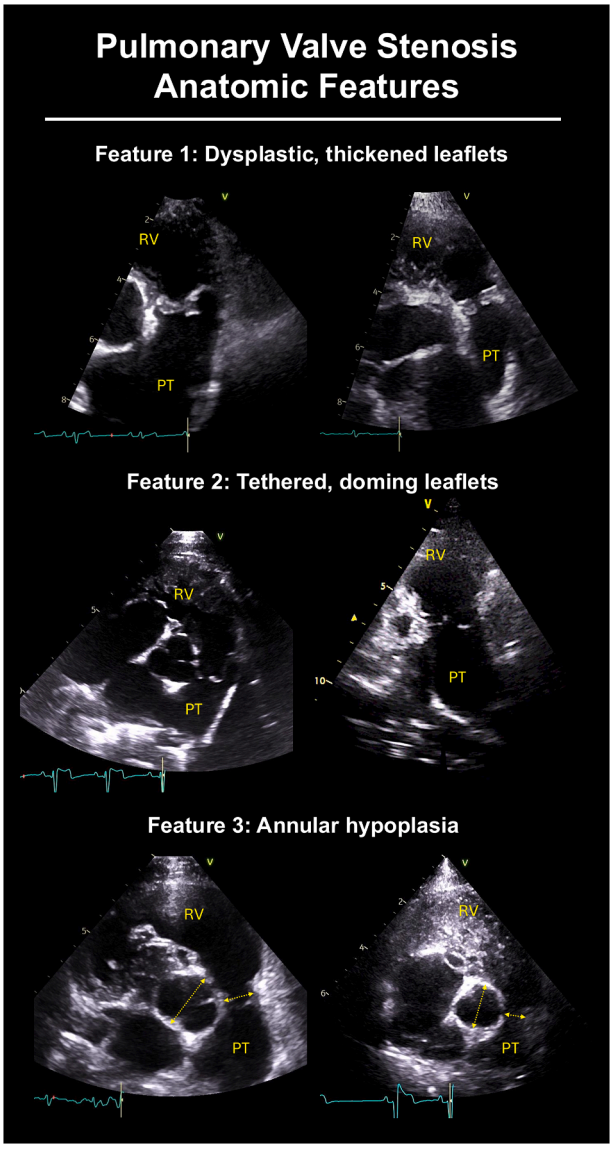


Figure 1 Example images demonstrating the features used for characterizing anatomic abnormalities associated with pulmonary valve stenosis in the present study. Two echocardiographic images are provided for each feature, with feature 1 representing dysplastic or thickened pulmonary valve leaflets, feature 2 representing tethered, doming pulmonary valve leaflets, and feature 3 representing annular hypoplasia of the pulmonary valve annulus. Dashed lines across the aorta and pulmonary valve annulus in the two example images of pulmonary annular hypoplasia allow for visual comparison of the size discrepancy between the two great vessels. PT: pulmonary trunk, RV: right ventricle.

subvalvular-dynamic = right ventricular hypertrophy related) (Fig. 2); and from a right parasternal short-axis view at the base of the heart, the following were obtained immediately after valve closure: aortic annulus diameter, pulmonary annulus diameter, and PT diameter midway between the valve annulus and bifurcation of the pulmonary artery branches. If images were not available for review, original measurements and subjective qualifiers provided in the echocardiogram report were recorded. Concurrent congenital abnormalities were also recorded.

Procedural data from the BVP procedure were collected, as available, and included the type and dimension of balloon catheter(s) and guidewire(s) used during the procedure, the angiographic balloon diameter to pulmonary valve annulus ratio, the total number of balloon inflations, total anesthesia time, and any reported procedural complications. Balloon catheters were characterized as high-pressure if inflated to $>$6 atm using an insufflator device or low-pressure if burst pressure was rated as $\leq$6 atm. Additionally, the severity of post-stenotic dilation of the PT was characterized based on pre-BVP angiograms, as available, as mild ($<$1.5 times greater than the pulmonary valve annulus), moderate (1.5–2 times greater than the pulmonary valve annulus), or severe ($>/=$ 2 times the diameter of the pulmonary valve annulus) as previously described [10]. Investigators also recorded whether a post-BVP angiogram was performed. In dogs that had a post-BVP angiogram, the presence of an angiographically evident PAD was recorded and further characterized as taking up $<$50% of the lumen of the PT or $>/=$ 50% of the lumen of the PT.

If periprocedural TEE was performed, investigators recorded whether the dissection was appreciated on this imaging modality and whether, at its maximum size on TEE, the dissection took up $<$50% of the lumen of the PT or $>/=$ 50% of the lumen of the PT. Investigators also recorded whether the PAD was visible on postoperative TTE imaging and, if so, in which views it could be seen (in a right parasternal short-axis view at the heart base only, the left cranial parasternal window only, or on both of these views), and whether the dissection took up $<$50% of the lumen of the PT or $>/=$ 50% of the lumen of the PT. In addition, maximum post-BVP transpulmonary velocity and corresponding pressure gradient were recorded. Based on the totality of imaging available for each case, investigators characterized the PAD as involving the PT only, the right pulmonary artery branch only, the left pulmonary artery branch only, or the PT with extension into the right or left pulmonary branch.

Follow-up data were collected for dogs as available and included whether a recheck echocardiogram was performed, whether any changes to the PAD were reported over time, whether another procedure was attempted for palliation,

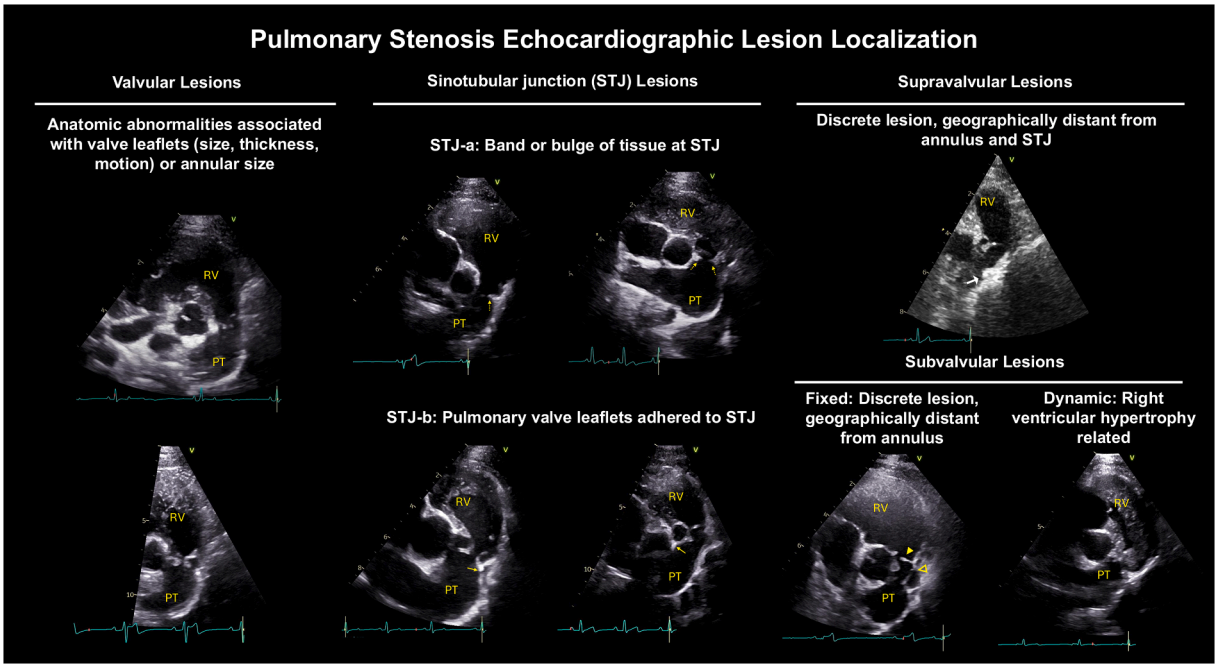


Figure 2 Example images demonstrating the echocardiographic lesion localization scheme employed in the present study for characterizing pulmonary valve stenosis location. Lesion localization included valvular lesions (encompassing anatomic abnormalities associated with pulmonary valve leaflet size, thickness, motion and/or annular size), sinotubular junction (STJ) lesions (subtype 'STJ-a' representing a band or bulge of tissue at the STJ and annotated in two example images with dashed yellow arrows and subtype 'STJ-b' representing pulmonary valve leaflets adhered to the STJ and annotated in two example images with solid yellow arrows), supravalvular lesions (representing discrete lesions that are geographically distant from the pulmonary valve annulus and STJ and annotated in the example image with a white arrow highlighting the supravalvular lesion) and subvalvular lesions (subtype 'subvalvular-fixed' representing a discrete lesion geographically distant from the pulmonary valve annulus and subtype 'subvalvular-dynamic' representing obstruction secondary to right ventricular hypertrophy). The 'subvalvular-fixed' subtype example image demonstrates a discrete subvalvular lesion (annotated with a solid yellow arrowhead) that is within the right ventricular outflow tract and geographically separate from the pulmonary valve annulus (annotated with an open yellow arrowhead), while the 'subvalvular-dynamic' subtype example image demonstrates severe right ventricular hypertrophy of the entire right ventricular outflow tract causing a dynamic subvalvular obstruction at end-systole. PT: pulmonary trunk, RV: right ventricle, STJ: sinotubular junction.

such as transpulmonary stenting, and current patient status as alive, deceased or unknown. Date and cause of death were recorded, when known, as were available necropsy findings in dogs that underwent postmortem examination. In dogs that were still alive or of unknown status, the last known date the dog was documented alive based on recheck examination or when follow-up with the owner or referring veterinarian was recorded.

Seven dogs with PAD development during BVP that have previously been described in the literature [9,10] were also included in this study, with additional data and subsequent follow-up information since previous publication collected.

Statistics

Descriptive statistics are reported as median (range) or proportion.

Results

Fifty-three dogs with PAD were identified across 19 veterinary referral centers. The searchable study period differed by center due to variable access to stored medical records, with a median search period of 13.5 years (range: 3.5—23.5 years). The median body weight of dogs diagnosed with PAD was 12.5 kg (range: 1.1—50.7 kg) and the median age was 0.8 years (range: 0.2—8.1 years). Thirty dogs were male (56.6%) and twenty-three were female (43.4%). The most represented breed was the mixed-breed dog (n = 17; 32.1%) followed by the French bulldog (n = 7,13.2%) and the Pit Bull terrier (n = 5, 9.4%). There were two dogs represented by each of the Labrador retriever, Chihuahua, whippet, and Yorkshire terrier breeds, and a single dog representing 16 different breeds (American bulldog, American Staffordshire terrier,

Australian heeler, Boykin spaniel, Bull terrier, bulldog, English bulldog, Golden retriever, Jack Russell rerrier, Pomeranian, Rhodesian ridgeback, shar pei, shih tzu, Siberian husky, West Highland White terrier and Wirehaired dachshund). Of the mixed breed dogs, six were specifically listed as Pit Bull terrier mixes, three as Chihuahua mixes and one as a Labrador retriever mix. done

At the time of the BVP procedure, 34 dogs (64.2%) had no reported clinical signs. The remaining 19 dogs (35.8%) had reported clinical signs including lethargy or exercise intolerance (n = 10), syncopal events (n = 8) and shortness of breath (n = 1). Five dogs (9.4%) had right-sided congestive heart failure documented pre-procedurally, one of which was incidentally diagnosed with no reported clinical signs. Regarding pre-procedural cardiac medications, there were nine dogs (17%) that were not receiving any cardiac medications at the time of the procedure. The remaining 44 dogs (83.0%) were receiving one, or a combination, of the following medications: atenolol (n = 37), furosemide (n = 5), an angiotensin converting enzyme inhibitor (n = 2), spironolactone (n = 1), diltiazem (n = 1), propranolol (n = 1) or pimobendan (n = 1).

Pre-operative TTE imaging provided a diagnosis of PS in all dogs, with 50 dogs (94.3%) considered to have a severe transpulmonary pressure gradient (exceeding 80 mmHg) and three dogs with a moderate transpulmonary pressure gradient (between 50 and 80 mmHg). The median transpulmonary pressure gradient in the study sample was 144 mmHg (range: 61–280 mmHg). Pulmonary valve anatomy features (Table 1) were characterized as solely dysplastic and thickened (n = 9), solely tethered/doming leaflets (n = 15), both dysplastic/thickened and tethered/doming leaflets (n = 17), both a hypoplastic annulus and tethered/doming leaflets (n = 1), and a combination of all three features (dysplastic/thickened leaflets, tethered leaflets and annular hypoplasia; n = 11). The echocardiographic pulmonary stenosis lesion localizations are presented in Table 1, with the most common characterization being valvular lesions only (n = 42). Secondary concentric hypertrophy of the right ventricle was characterized as mild (n = 2), moderate (n = 8) and severe (n = 43), while subjective right atrial dilation was characterized as none (n = 1), mild (n = 14), moderate (n = 25) and severe (n = 13). Concurrent congenital cardiac disease was present in 17 dogs (32.1%) and included patent foramen ovale (n = 8), tricuspid valve dysplasia (n = 5), patent ductus arteriosus (n = 2), a small, restrictive ventricular septal defect and patent foramen ovale (n = 1), and tricuspid valve dysplasia and patent foramen ovale (n = 1).

Table 1 Distribution of pulmonary valve anatomy features and echocardiographic pulmonary stenosis lesion localization in 53 dogs with pulmonary valve stenosis and pulmonary artery dissection during pulmonary balloon valvuloplasty.

Pulmonary valve anatomy features	Number of dogs
Feature 1 only: dysplastic, thickened leaflets	N = 9
Feature 2 only: tethered or doming leaflets	N = 15
Feature 3 only: annular hypoplasia	N = 0
Combined features 1 and 2	N = 17
Combined features 2 and 3	N = 1
Combined features 1, 2 and 3	N = 11
Echocardiographic pulmonary stenosis lesion localization	**Number of dogs**
Valvular only	N = 42
STJ-a only	N = 1
STJ-b only	N = 1
Supravalvular only	N = 0
Subvalvular-fixed only	N = 0
Subvalvular-dynamic only	N = 0
Combined valvular and subvalvular-dynamic	N = 2
Combined valvular and STJ-b	N = 5
Combined valvular, STJ-b and subvalvular-dynamic	N = 2

Angiographic post-stenotic dilation of the PT in the dogs of this study was mild (n = 5), moderate (n = 22), severe (n = 16), or uncharacterized (n = 10). Angiographic measurements of the pulmonary valve annulus were available in 49 dogs. Of these 49 dogs, the median angiographic balloon diameter to pulmonary valve annulus ratio of the largest balloon that was utilized during the BVP was 1.4 (range: 1.0–1.58). Forty-five dogs had a single balloon catheter size utilized during their BVP procedure. Six dogs had more than one balloon catheter used, with four of these dogs having two balloon sizes used and two of them having three different balloon sizes used. Two dogs had a double BVP procedure with two balloons inflated simultaneously. High-pressure balloons inflated with an insufflator and having a rated burst pressure >6 atm were utilized in 27 dogs (50.9%). In four dogs, a Z-med balloon catheter was used in which two different burst pressure ratings (5 or 8 atm) are available for purchase in the balloon catheter dimension described for the case. While these balloons were inflated with an insufflator, it is unclear whether the rated burst pressure was 5 or 8 atm, so these dogs were not characterized as

definitively high-pressure or low-pressure. The remaining 22 dogs (41.5%) had a low-pressure balloon used for their BVP. Dogs that had both a low-pressure and a high-pressure balloon used during the BVP (n = 4) were included in the high-pressure category. The frequency of high- and low-pressure balloon usage in association with PAD over the study period is presented graphically in Figure 3. The number of balloon inflations during the BVP procedure was recorded in 47 dogs, with a median value of three inflations (range: 1–27). Rupture of a balloon catheter during the BVP procedure was reported in four dogs (7.5%), with one of these balloons having a pinpoint rupture near the site of the subsequently recognized PAD rather than a longitudinal balloon rupture. The stiffness of the stiffest guidewire used during the BVP was characterized as regular (n = 19), stiff/intermediate (n = 2), extra-stiff (n = 23), super-stiff (n = 8), or ultra-stiff (n = 1).

A post-BVP angiogram was not performed in 22 dogs (41.5%). In the 31 dogs in which a post-BVP angiogram was performed, the PAD was visible in 28/31 dogs (90.3%) and was considered to take up <50% of the PT lumen at its maximum size in 19 dogs, >/ = 50% of the PT lumen in three dogs, only affected the right pulmonary artery branch in one dog, and was uncharacterized in five dogs. Pre- and post-BVP angiograms in three dogs are provided in Figure 4.

Perioperative TEE imaging was performed in 28 dogs (52.8%) but was pre-BVP only in three of these dogs and thus unable to diagnose a PAD that occurred during BVP. The PAD was visible on TEE

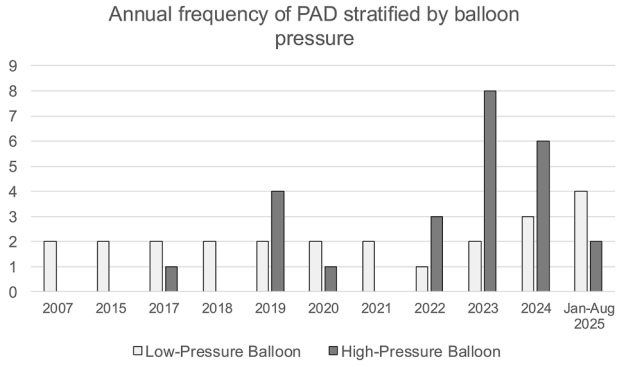


Figure 3 Graphical representation of the annual frequency of a pulmonary artery dissection (PAD) complication developing after a balloon valvuloplasty procedure in dogs with pulmonary valve stenosis. Dogs that experienced PAD are stratified based upon whether a low-pressure (light gray) or high-pressure (dark gray) balloon was utilized for the balloon valvuloplasty procedure. If both a low- and a high-pressure balloon were used during the balloon valvuloplasty procedure, the dog was categorized under high-pressure balloon.

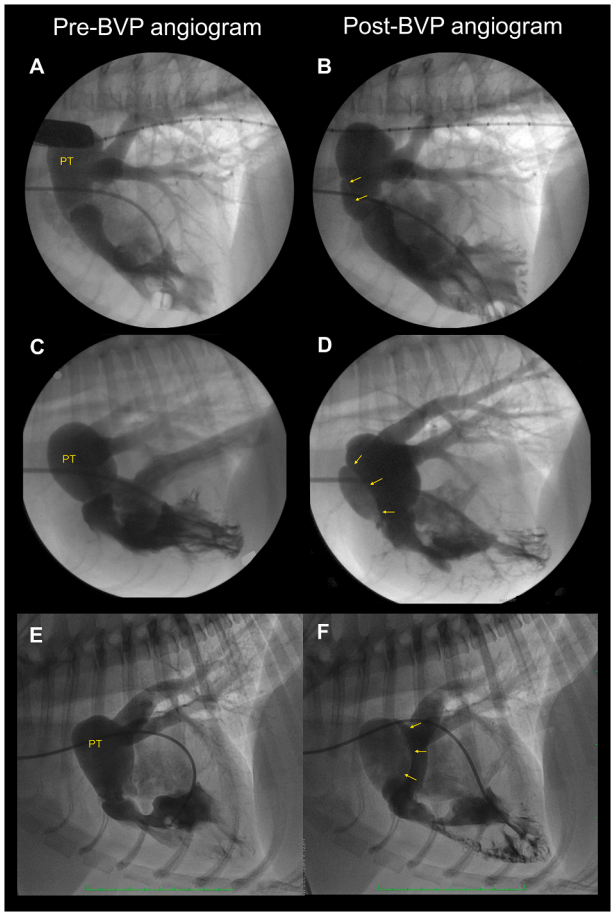


Figure 4 Example pre- and postprocedure angiographic images from dogs with pulmonary valve stenosis that developed a pulmonary artery dissection (PAD) after a balloon valvuloplasty (BVP) procedure. Dog one (image panels A and B) and dog two (image panels C and D) developed a PAD post-BVP that occupied <50% of the pulmonary trunk (PT) lumen, while dog three (image panels E and F) developed a PAD post-BVP that occupied >50% of the PT lumen. The dissecting membrane of the PAD is highlighted with yellow arrows in each post-procedure angiographic image.

imaging post-BVP in 21/25 dogs (84%) and was considered to occupy <50% of the lumen of the PT at its maximum size in ten dogs, >/ = 50% of the PT lumen in ten dogs, and in one dog, the amount of the lumen occupied by the dissection varied above and below 50% based on the individual image reviewed. Only 11 dogs had both angiographic and TEE characterization of the amount of the PT lumen occupied by the PAD, with 9/11 agreeing across imaging modalities. In the two dogs where angiography and TEE did not agree, one dog had an angiographic assessment of <50% of the PT lumen being occupied by the PAD while the TEE assessment was >/ = 50% of the PT lumen. The other dog with disagreement was the one in

which the TEE assessment varied above or below 50% of the PT lumen occupied by the dissection based upon which image was assessed. In this dog the angiographic assessment was <50% of the PT lumen occupied by the dissection.

A visible PAD was reported to be clearly appreciated on postoperative TTE imaging after the BVP procedure in 33 of 53 dogs (62.2%), though in two of these dogs the PAD was not appreciated during the TTE study performed 24 h postoperatively and was only appreciated on a TTE study performed 21-days postoperatively and 1 month postoperatively, respectively. In an additional four dogs, the PAD was suspected but difficult to confirm on TTE imaging. In nine dogs the PAD was not appreciated on TTE imaging, and in seven dogs this variable was unrecorded. An example of all post-BVP imaging modalities in a single dog are shown in Figure 5, in which there is a disparity in recognition of the PAD on TTE compared to angiography and TEE. Cine loops of TEE and TTE images for the same dog are provided in Videos 1 and 2. Post procedural pressure gradient obtained on post operative TTE imaging was considered mild (<50 mmHg) in 27 dogs, moderate (50–80 mmHg) in 15 dogs, and remained severe (>80 mmHg) in seven dogs.

The diagnosis of PAD was made on post-BVP TTE in 10 dogs (18.9%), a post-BVP angiogram in 14 dogs (26.4%), post-BVP TEE in 11 dogs (20.8%), both post-BVP angiogram and TEE in 12 dogs (22.6%), both post-BVP angiogram and TTE in three dogs (5.7%), necropsy in two dogs (3.8%) and with cardiac computed tomography imaging in one dog (1.9%). In four of the dogs, the diagnosis was made incidentally during pre-operative imaging (TEE, angiography, or computed tomography) for transpulmonary valve stent placement (Fig. 6). Given the totality of available imaging on each case, the

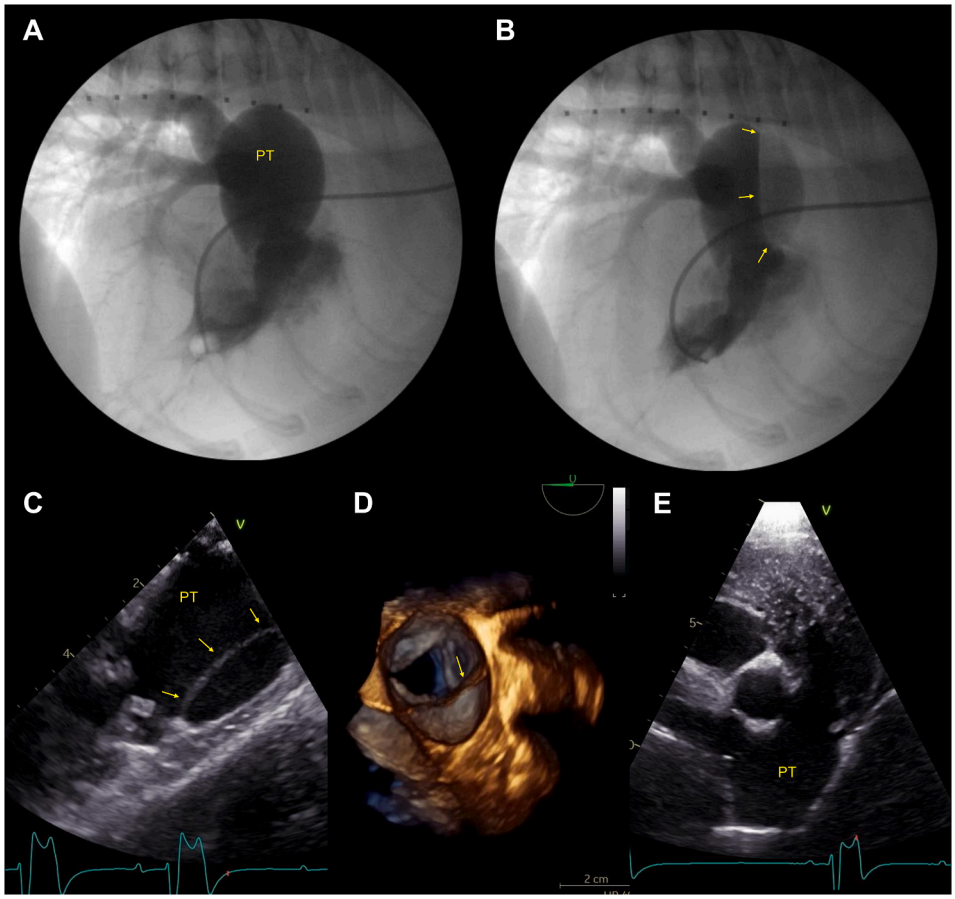


Figure 5 A collection of images from a single dog with pulmonary valve stenosis that developed a pulmonary artery dissection (PAD) after a balloon valvuloplasty (BVP) procedure. Image A and B are pre- and post-BVP angiographic images, with the dissecting membrane of the PAD highlighted with yellow arrow in image B. Image C and D are 2-dimensional and 3-dimensional transesophageal echocardiographic images of the pulmonary valve and pulmonary trunk (PT) obtained post-BVP. The dissecting membrane of the PAD is identified with yellow arrows in each image. Image E is a 24-h post-BVP transthoracic echocardiographic short axis image at the heart base optimized for the PT. In contrast to the post-BVP angiogram and transesophageal echocardiographic images, the dissecting membrane of the PAD is not readily identifiable on the post-BVP transthoracic echocardiographic image.

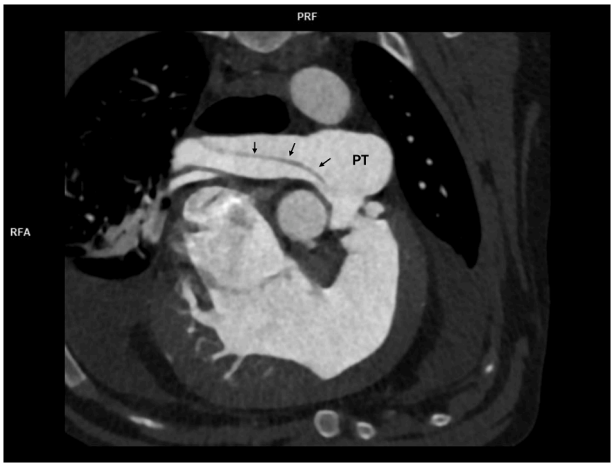


Figure 6 A cardiac computed tomography image of a dog with pulmonary valve stenosis that developed a pulmonary artery dissection (PAD) after balloon valvuloplasty. A dissecting membrane is identified with black arrows and extends from the sinotubular junction to the right pulmonary artery. PT: pulmonary trunk.

PAD was characterized as involving the PT only (n = 41), the PT with extension into the right pulmonary artery branch (n = 8), the PT with extension into the left pulmonary artery branch (n = 3), and the right pulmonary artery branch only (n = 1).

Three dogs experienced cardiopulmonary arrest during the BVP procedure and died. Two additional dogs had perioperative deaths shortly after BVP, with one dog euthanized due to severe, unresponsive hypoxemia during anesthetic recovery and the other dying spontaneously two hours post-BVP using a double balloon technique. Both perioperative deaths were assumed to be related to clinical decompensation secondary to the PAD. Of these five dogs, one dog had a post-BVP angiogram and perioperative TEE, with both imaging modalities assessing that the PAD occupied >50% of the PT lumen, and one dog had perioperative TEE imaging that assessed the PAD to occupy >50% of the PT lumen. The other three dogs had no perioperative imaging to assess the relative size of the PAD. Four of these five dogs had a necropsy performed which confirmed the PAD, three of which were described in a previous publication [10]. The remaining 48 dogs (90.6%) survived the perioperative period and were discharged from the hospital.

With regard to long-term follow-up, 39/48 dogs (81.3%) had a follow-up examination or other contact after discharge from their BVP procedure when they were confirmed to be alive. The median time from BVP to the date of last follow-up (or subsequent date of death, if applicable) was 2.0 years (range: 0.2–10.5 years). There were 27/39 dogs (69.2%) that had one or more year of follow-up available, all of whom were documented to be alive one year after experiencing PAD. Follow-up TTE studies were performed in 38 dogs at some point after discharge from their BVP procedure, with the appearance of the PAD described as improved (n = 2), unchanged (n = 20), not visible or poorly defined (n = 7), or not characterized (n = 8). In one dog, the appearance of the PAD on a follow-up TTE performed three months post-BVP seemed larger than on the initial TEE at the time of BVP, however, no immediate postoperative TTE image was available with which to compare.

Four dogs with long-term follow-up had died at the time of data collection. One dog was euthanized 10.5 years after BVP due to hepatic disease. A second dog was euthanized one year and one day after BVP due to endocarditis of the pulmonary valve. On necropsy of this dog there was necrotizing valvular endocarditis of the pulmonary valve with intralesional bacteria, moderate and multifocal hemorrhage of the adventitia of the pulmonary artery, and necrohemorrhagic suppurative pneumonia. An obvious PAD was not reported by the attending pathologist at the time of necropsy. The third dog was a French Bulldog that reportedly died at home three years after BVP under unknown circumstances. This dog would have been four years of age at the time of death. The fourth and final dog was euthanized due to progressive right-sided congestive heart failure 2.5 years after BVP.

Six dogs went on to have placement of a transpulmonary stent at 4 months, 8 months, 10 months, 14 months, and two at 2 years post BVP, respectively. In all cases, placement of the stent was not specifically to address the PAD, but rather to address the underlying severe PS, with a possibility that increased stability of the PAD could be achieved with stent placement. A covered stent was placed in one of the six dogs, while the remainder had biliary stents of variable sizes placed. No complications associated with worsening of the PAD occurred during stent placement and all six dogs survived the procedure.

Discussion

In dogs that experienced PAD during a BVP intervention in this study, immediate postoperative survival was high, with 90.6% of dogs surviving to discharge. This was far greater immediate survival than the previously published 50.0% survival rate from a case series of six dogs [10], and nearly as high as the published survival rate for the BVP procedure in dogs of 95.7% [15]. Long-term survival

after PAD also appears to be high, with all dogs in which one year or more of follow-up data was available still alive at that time point. The mortality rate associated with PAD in humans has historically been considered very high, with emergent surgical intervention typically considered the treatment of choice [1—5]. In more recent years, however, advanced imaging modalities are detecting more chronic, incidental PAD cases, typically in the setting of chronic pulmonary arterial hypertension [16,17]. Some instances of longer-term survival with medical management alone are now reported in these settings [16,17]. Additionally, case reports exist of PAD development during BVP intervention in children in which the dissection remained static and asymptomatic for many years postprocedure without surgical intervention [12,13].

In one dog, the development of infectious endocarditis of the pulmonary valve after BVP and PAD could plausibly be considered a long-term complication of the original dissection due to presumed altered blood flow and endothelial disruption adjacent to or directly affecting the pulmonary valve. Evidence of an active PAD was not reported at the time of necropsy, however, suggesting that spontaneous healing may have occurred in the intervening year and making a definitive cause—effect relationship difficult to establish. Additionally, in the French Bulldog that died at home three years after PAD secondary to BVP, details surrounding the death were not available and a necropsy was not performed. The cause of spontaneous death in this dog could have been related to rupture of the PAD or been non-cardiac in origin.

Regarding demographic variables in dogs experiencing PAD, male dogs were slightly over-represented (56.6%) in the present study sample compared to females. This is similar to humans, in which the male:female ratio of PAD cases is reported as 1.1:1 [5]. Sex-based differences in vascular smooth muscle cell prevalence, endothelial cell populations, artery size, arterial stiffness, and the prevalence of comorbid conditions associated with arterial dissection (*e.g.* connective tissue disorders, pulmonary hypertension) are some of the many factors that appear to play a role in sex-based predispositions for arterial dissections in humans, with these sex-related differences varying based on the age of the patient and the vessel involved [18]. For example, pulmonary artery dissections are less sexually dimorphic than aortic dissections in humans, with males accounting for 67.5% of aortic dissection cases [18]. Whether any of the sex-based predispositions for arterial dissection appreciated in humans apply to canine populations remains unknown.

From a breed perspective, mixed breed dogs were the most represented 'breed' in this study. Looking across the entire population, however, breeds of similar heritage appeared to be frequently represented. The American Kennel Club recognizes seven breeds as having common heritage and characteristics that allow them to be grouped together under the label of 'Bully breeds,' including the American Staffordshire terrier, the Bulldog, the Bull Mastiff, the Bull terrier, the French Bulldog, the Miniature Bull terrier, and the Staffordshire Bull terrier. The 'Pit Bull terrier,' however, is not a recognized breed by the American Kennel Club as it is seen as a general term for dogs with similar physical traits, largely overlapping with those of the so-called 'Bully breeds.' For purposes of this study, if dogs characterized as Pit Bull terriers, Pit Bull terrier mixes, and dogs of the recognized 'Bully breeds' are combined, they represent 23/53 dogs of this study (43.4%). Whether this speaks to a true breed predisposition for PAD or simply to an underlying predisposition for PS across these breeds is difficult to discern with the available data.

Humans with connective tissue disorders like vascular Ehlers-Danlos syndrome [19] or Marfan syndrome are predisposed to arterial dissection or rupture due to deficient or structurally abnormal collagen or fibrillin, respectively. These syndromes are rare in veterinary medicine, with no single breed clearly over-represented. Of note, however, a single case report of a 'Marfan-like' syndrome in an English bulldog with a left subclavian artery dissection and connective tissue abnormalities has been published [20]. Additionally, Pit Bull terriers [21,22] and Alapaha Blue Blood Bulldogs [22] are among the reported breeds with genetic mutations and associated disease phenotypes of Ehlers-Danlos syndrome, though the dogs in these studies had no reported vascular manifestations of the disease. Whether an underlying connective tissue disorder could be contributory to the risk of PAD in the 'Bully breeds' of this study remains unclear and warrants further investigation.

The median age of dogs experiencing PAD in this study sample was 0.8 years, with a range of 0.2—8.1 years. Advanced age prior to BVP could plausibly be considered a risk factor for increased PAD risk due to increased chronicity of a high velocity jet causing trauma to the PT. The present data do not appear to support that premise, however, as the median age in this study is very similar to that the median age of 0.75 years reported in two previous studies [15,23] describing complication rates in a total of

147 dogs with PS that underwent BVP, none of which were known to have developed a PAD.

Another plausible risk factor for PAD is use of an over-sized balloon that stretches the PA excessively and leads to vascular damage. Historically, it has been recommended that a balloon catheter that is 1.2—1.5 times the width of the angiographically measured pulmonary valve annulus be selected for BVP procedures [24,25], including those performed with high-pressure balloons [26]. In the present study sample, only three dogs had an angiographic balloon to annulus ratio that exceeded 1.5, while the median ratio for this cohort was 1.4. This suggests that balloon oversize, at least as assessed relative to the angiographically measured diameter of the pulmonary valve annulus, was not a leading contributor to PAD. It is important to recognize the inherent limitations of relying on a single imaging modality to assess an accurate pulmonary valve annulus diameter, however, as angiographic resolution and adequate alignment with a true sagittal plane across the pulmonary valve annulus may vary from dog to dog. In some dogs, other imaging modalities such as TTE, TEE or computed tomography may produce better image quality and variable annular dimensions compared to angiography. Ideally, a multi-modality imaging approach should be utilized, with the imaging modality having the best image resolution of the pulmonary annulus relied upon for balloon selection and sizing. Given the retrospective nature of this study, a comparison of pulmonary valve annulus diameters across various imaging modalities was not feasible.

Balloon valvuloplasty procedures can be performed with a variety of different balloon catheters. Use of high-pressure balloon catheters in veterinary medicine has been reported as safe and effective [26], though a recent randomized, prospective, matched-pair clinical trial did not find statistical differences in the reduction of PS severity post-BVP in dogs treated with high pressure versus low pressure BVP interventions [27]. Importantly, there was also no evidence of increased myocardial injury or complication rate with high-pressure balloons versus low-pressure balloons in that trial [27]. The results of this retrospective study suggest that the rare complication of PAD also occurs relatively equally across both high- and low-pressure balloon BVP procedures, with 50.9% of the present cases having been ballooned with a high-pressure balloon. When interpreting this data, however, it is important to consider changing practice habits over time. While low-pressure balloons have been routinely utilized for the BVP procedure since the procedure became standard of care in veterinary cardiology, high-pressure balloons have only gained more widespread use in recent years. Without knowledge of the total number of high-versus low-pressure BVP procedures performed per year at each study site, truly assessing the frequency of PAD as it relates to balloon type remains challenging and requires further study.

From a morphology standpoint, 51/53 dogs (96.2%) included in the present study were characterized as having valvular PS, with various combinations of tethered leaflets, thickened/dysplastic leaflets, or annular hypoplasia features. Seven dogs were characterized as having STJ lesions in addition to valvular lesions of the leaflets and/or annulus, while two dogs were characterized has having solely STJ lesions. Because characterization of supravalvular PS is not standardized across veterinary cardiologists, with some considering lesions associated with the STJ to fall under the umbrella of valvular PS while others consider similar lesions to represent supravalvular PS, specific anatomic descriptions were included within the study methods to ensure that lesions were similarly grouped, and STJ lesions were grouped on their own. In this cohort of dogs, no dogs had supravalvular stenosis defined as a geographically distant lesion above the level of the STJ and pulmonary valve annulus. Given the spectrum of annular, leaflet and STJ lesions represented in this cohort, no specific conclusions about risk of PAD and lesion morphology can be drawn. In two dogs, a concurrent patent ductus arteriosus was present in addition to PS. Given that PAD is a potential complication reported in dogs with patent ductus arteriosus [6,7], this combination of congenital heart disease may have contributed to increased risk of PAD development during BVP in these specific dogs.

Finally, post-BVP angiograms were not performed in 41.5% of dogs, while post-BVP TEE imaging was not performed in 52.8% of dogs. Dogs in which no immediate post-BVP imaging was performed may be at increased risk of having a PAD complication go unnoticed, as a visible PAD was only clearly appreciated in 62.2% of dogs in this study on postoperative TTE imaging, while 90.3% and 84% were reportedly visible on angiography and TEE imaging, respectively.

Limitations

Limitations of this study largely relate to its retrospective nature. Unfortunately, given the apparent rarity of PAD development during BVP, long search periods across numerous centers were

needed to accumulate a meaningful number of cases. Long search periods often encompassed a variety of changes in medical record systems across centers. This proved challenging for identifying the total number of PS cases undergoing BVP during the study period at all participating institutions for purposes of calculating the frequency of a PAD complication. Similarly, the total number of dogs of individual breeds undergoing BVP at each participating institution over the study period could not reasonably be assessed to better determine possible breed-related PAD predispositions versus breed-related predispositions to PS in general. Additionally, since cases were identified retrospectively through participating centers, selection bias toward memorable or clinically recognized PAD cases cannot be excluded. Lastly, given the retrospective nature of the study, different dogs had different pre-procedural imaging modalities performed that were relied upon to confirm lack of a PAD prior to the BVP procedure. While all dogs had pre-procedural TTE and pre-procedural angiography performed, not all dogs had pre-procedural TEE or other advanced imaging modalities. Given the variable sensitivity of different imaging modalities at detecting PAD in different dogs, we cannot exclude the possibility that a pre-existing dissection was missed on initial imaging.

Conclusions

In conclusion, dogs experiencing PAD during BVP have a high likelihood of survival in the immediate and long-term without subsequent intervention. There is a potential over-representation of Pit Bull terriers and 'Bully' breeds. Age at time of BVP, angiographically determined balloon to annulus ratio, and high-pressure versus low-pressure BVP were not determined to be obvious risk factors for PAD development during BVP intervention. Immediate post-BVP imaging with angiography or TEE is recommended for improved diagnosis of PAD as TTE is less sensitive for adequate visualization of dissecting membranes.

Ethics declaration

This study was conducted in accordance with the following guidelines for animal welfare and/or reporting: An IACUC protocol was not required for a retrospective study with medical record review only. Ethics committee approval was not required. This was a retrospective study that collected data from medical records on a standard of care procedure that was performed to treat underlying congenital heart disease in dogs.

Declaration of Competing Interest

The authors do not have any conflicts of interest to disclose.

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Supplementary data

Supplementary data associated with this article can be found, in the online version, at https://doi.org/10.1016/j.jvc.2026.09.007.

Video	Title	Description
1	Perioperative transesophageal echocardiography (TEE) 2-dimensional and 3-dimensional images post-balloon valvuloplasty	Video 1 rotates through a biplane, 2-dimensional TEE view of the pulmonary valve and pulmonary trunk with a pulmonary artery dissection visible in both planes, followed by a single plane, 2-dimensional image with simultaneous color flow Doppler over the dissecting membrane to demonstrate the difference in blood flow within the pulmonary artery compared to the false lumen, and finally a 3-dimensional image looking from the pulmonary trunk towards the pulmonary valve in which the dissecting membrane is also clearly visible.
2	Transthoracic echocardiography (TTE) 2-dimensional images of the same dog 24-hours post-balloon valvuloplasty	Video 2 shows a right parasternal short axis TTE view at the level of the heart base in which the pulmonary trunk is optimized, followed by a left-sided TTE view also optimized for the pulmonary valve and pulmonary trunk. A dissecting membrane is difficult to definitively discern in the TTE images.

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